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SQL Mathematics & Number Notes

Integer vs Floating Point

Integer

25

Exact value.

Floating Point

25.6789

Approximate representation.

Exact Types

- DECIMAL
- NUMERIC

Ideal for: Money, Taxes, Banking , DECIMAL(8,2) 8 is total digits=precision and 2 is after decimal digit known as scale

Type Conversion

Implicit Conversion

SQL converts automatically.

Example:

```
SELECT 10 + 5.5;
```

Result: 15.5

Explicit Conversion

Using CAST

```
SELECT CAST(12.9 AS INT);
```

NULL Behavior

NULL propagates.

```
SELECT 5 + NULL;
```

↓ **NULL**

```
ROUND(NULL)
```

↓ **NULL**

ROUND()

ROUND() rounds a number to the nearest value.

ROUND(number)

or

ROUND(number, decimal_places)

```
SELECT ROUND(12.3456, 2);
```

↓ **12.35**

Round to Integer

```
SELECT ROUND(12.7);
```

↓ **13**

```
SELECT ROUND(12.2);
```

Negative Precision

Very common interview topic.

Round to nearest ten

```
SELECT ROUND(456, -1);
```

↓ **460**

Nearest hundred

```
SELECT ROUND(456, -2);
```

↓ **500**

Nearest thousand

```
SELECT ROUND(6789, -3);
```

↓ **7000**

ROUND vs TRUNCATE

Suppose 12.987

ROUND

↓ **12.99**

ROUND performs mathematical rounding.

TRUNCATE

↓ **12.98**

TRUNCATE simply removes digits.

CEIL() / CEILING()

3.1 Introduction

CEIL() always rounds up toward positive infinity.

Even if the decimal part is very small.

Example:

```
12.01
```

↓ 13

3.2 Syntax

```
CEIL(number)
```

Positive Numbers

```
SELECT CEIL(12.1);
```

↓ 13

Already integer

```
SELECT CEIL(20);
```

↓ 20

Negative Numbers

Very important.

```
SELECT CEIL(-4.7);
```

↓ -4

Because -4 is greater than -4.7.

FLOOR()

4.1 Introduction

FLOOR() always rounds down toward negative infinity.

Example:

```
12.99
```

↓ 12

4.2 Syntax

```
FLOOR(number)
```

Negative Numbers

Interview favorite.

```
SELECT FLOOR(-4.2);
```

↓ -5

Syntax (Power)

```
POWER(base, exponent)
```

or

```
POW(base, exponent)
```

(MySQL also supports POW().)

Fractional Powers

Fractional exponents represent roots.

Square root

$$x^{1/2} = x^{1/2} = \sqrt{x} \quad x^{1/2} = x^{1/2}$$

Example

```
SELECT POWER(25, 0.5);
```

↓ 5

Cube root

```
SELECT POWER(27,1.0/3);
```

↓ 3

Population Growth

Population	100000
Growth rate	5%
Years	10

```
SELECT 100000 * POWER(1.05,10);
```